

Delegation in Strategic Environments and Equilibrium Uniqueness

Fedor Sandomirskiy - Princeton

Ben Wincelberg - Caltech

PSE, May 1, 2026

Motivation

- **coordination** drives familiar market failures: cartels, bidding rings, tacit collusion

Motivation

- **coordination** drives familiar market failures: cartels, bidding rings, tacit collusion
- new channel: **algorithmic delegation**
 - ▶ participants authorize an intermediary (a platform or AI agents) to act for them, coordinating at scale
 - ▶ collusive outcomes even when each runs its own algorithm ([Calvano, Calzolari, Denicolo, and Pastorello, 2020](#); [Klein, 2021](#); [Fish, Gonczarowski, and Shorrer, 2024](#); [Assad, Clark, Ershov, and Xu, 2024](#))

Motivation

- **coordination** drives familiar market failures: cartels, bidding rings, tacit collusion
- new channel: **algorithmic delegation**
 - ▶ participants authorize an intermediary (a platform or AI agents) to act for them, coordinating at scale
 - ▶ collusive outcomes even when each runs its own algorithm ([Calvano, Calzolari, Denicolo, and Pastorello, 2020](#); [Klein, 2021](#); [Fish, Gonczarowski, and Shorrer, 2024](#); [Assad, Clark, Ershov, and Xu, 2024](#))
- growing **antitrust concerns**
 - ▶ US DOJ (2025): “we are on the verge of autonomous algorithmic collusion”
 - ▶ EC Horizontal Guidelines (2023): algorithms as “a device to facilitate collusion (collusion by code)”
 - ▶ UK CMA (2026): “detering algorithmic collusion” is a priority

Motivation

- **coordination** drives familiar market failures: cartels, bidding rings, tacit collusion
- new channel: **algorithmic delegation**
 - ▶ participants authorize an intermediary (a platform or AI agents) to act for them, coordinating at scale
 - ▶ collusive outcomes even when each runs its own algorithm (Calvano, Calzolari, Denicolo, and Pastorello, 2020; Klein, 2021; Fish, Gonczarowski, and Shorrer, 2024; Assad, Clark, Ershov, and Xu, 2024)
- growing **antitrust concerns**
 - ▶ US DOJ (2025): “we are on the verge of autonomous algorithmic collusion”
 - ▶ EC Horizontal Guidelines (2023): algorithms as “a device to facilitate collusion (collusion by code)”
 - ▶ UK CMA (2026): “detering algorithmic collusion” is a priority
- state of the art: the scope is **poorly understood** (FTC, 2025)
 - ▶ depends on the strategic environment, the algorithm designer, implementation, and the AI black box

Motivation

- **coordination** drives familiar market failures: cartels, bidding rings, tacit collusion
- new channel: **algorithmic delegation**
 - ▶ participants authorize an intermediary (a platform or AI agents) to act for them, coordinating at scale
 - ▶ collusive outcomes even when each runs its own algorithm (Calvano, Calzolari, Denicolo, and Pastorello, 2020; Klein, 2021; Fish, Gonczarowski, and Shorrer, 2024; Assad, Clark, Ershov, and Xu, 2024)
- growing **antitrust concerns**
 - ▶ US DOJ (2025): “we are on the verge of autonomous algorithmic collusion”
 - ▶ EC Horizontal Guidelines (2023): algorithms as “a device to facilitate collusion (collusion by code)”
 - ▶ UK CMA (2026): “detering algorithmic collusion” is a priority
- state of the art: the scope is **poorly understood** (FTC, 2025)
 - ▶ depends on the strategic environment, the algorithm designer, implementation, and the AI black box

Our paper: strategic environments where the concerns are moot, regardless of the intermediary details

Our contribution

Our question

When does a strategic interaction deliver a **unique equilibrium outcome** even if players can [delegate](#) play to an intermediary with substantial coordinating power?

Our contribution

Our question

When does a strategic interaction deliver a **unique equilibrium outcome** even if players can **delegate** play to an intermediary with substantial coordinating power?

- a delegation outcome is a **distribution over action profiles**
- we ask which ones an intermediary can sustain so that no agent wants to **opt out**

Our contribution

Our question

When does a strategic interaction deliver a **unique equilibrium outcome** even if players can **delegate** play to an intermediary with substantial coordinating power?

- a delegation outcome is a **distribution over action profiles**
- we ask which ones an intermediary can sustain so that no agent wants to **opt out**
- two extremes of intermediary power:
 - ▶ **weak**: cannot punish deviators $\Rightarrow$ **coarse correlated equilibrium** (CCE)
 - ▶ **strong**: can coordinate adversarial punishments $\Rightarrow$ **individually rational correlated profiles** (IRCP)

Our contribution

Our question

When does a strategic interaction deliver a **unique equilibrium outcome** even if players can **delegate** play to an intermediary with substantial coordinating power?

- a delegation outcome is a **distribution over action profiles**
- we ask which ones an intermediary can sustain so that no agent wants to **opt out**
- two extremes of intermediary power:
 - ▶ **weak**: cannot punish deviators $\Rightarrow$ **coarse correlated equilibrium** (CCE)
 - ▶ **strong**: can coordinate adversarial punishments $\Rightarrow$ **individually rational correlated profiles** (IRCP)

Main contribution: characterizations of games with a unique CCE or a unique IRCP

- uniqueness reduces to a **negative-sum externality** condition
- less demanding than expected, tractable, and suitable for collusion-proof design
- $\Rightarrow$ new sufficient conditions for uniqueness of NE and CE

Related literature

- **Correlated solution concepts:** CE (Aumann, 1974, 1987); CCE (Hannan, 1957; Moulin and Vial, 1978); IRCP (Myerson, 1991, Chapter 6)
- **Foundations:** communication $\rightarrow$ CE (Bárány, 1992; Ben-Porath, 1998; Gerardi, 2004; Lehrer and Sorin, 1997); regret learning $\rightarrow$ CCE, CE (Hannan, 1957; Foster and Vohra, 1997; Fudenberg and Levine, 1999; Hart and Mas-Colell, 2000); folk theorem logic $\rightarrow$ IRCP (Benoit, Krishna, et al., 1984; Fudenberg and Maskin, 1986; Aumann and Shapley, 1976);
- **Unique CE:** convex potential (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025); zero-sum (Forges, 1990; Viossat, 2006; Kattwinkel, Niemeyer, Preusser, and Winter, 2022; Kartik, Squintani, and Tinn, 2024); auctions (Feldman, Lucier, and Nisan, 2016)
- **Unique CCE:** strict social concavity (Even-Dar, Mansour, and Nadav, 2009; Hart and Mas-Colell, 2015); non-atomic congestion (Koessler et al., 2025); other examples (Nadav and Piliouras, 2010; Baudin, Scarsini, and Venel, 2023; Ahunbay and Bichler, 2025)

Roadmap

- ① Model & solution concepts: NE, CE, CCE, IRCP
- ② Unique IRCP
- ③ Unique pure CCE
- ④ Application: contests and potential games
- ⑤ Unique mixed CCE
- ⑥ Extensions and open problems

Model & solution concepts

Delegation as a correlated device

- finite normal-form game $G = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$

Delegation as a correlated device

- finite normal-form game $G = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$
- an **intermediary** selects a distribution $\mu \in \Delta(A)$ and implements $a \sim \mu$ on players' behalf

Delegation as a correlated device

- finite normal-form game $G = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$
- an **intermediary** selects a distribution $\mu \in \Delta(A)$ and implements $a \sim \mu$ on players' behalf
- player i can **opt out of delegation** by switching ex ante to a fixed action $a'_i \in A_i$

Delegation as a correlated device

- finite normal-form game $G = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$
- an **intermediary** selects a distribution $\mu \in \Delta(A)$ and implements $a \sim \mu$ on players' behalf
- player i can **opt out of delegation** by switching ex ante to a fixed action $a'_i \in A_i$

Key modeling choice: what can the intermediary do **after** an opt-out?

Delegation as a correlated device

- finite normal-form game $G = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$
- an **intermediary** selects a distribution $\mu \in \Delta(A)$ and implements $a \sim \mu$ on players' behalf
- player i can **opt out of delegation** by switching ex ante to a fixed action $a'_i \in A_i$

Key modeling choice: what can the intermediary do **after** an opt-out?

Two extremes

weak intermediary: cannot punish vs **strong intermediary:** can coordinate adversarial punishments

CCE : weak intermediary

Coarse Correlated Equilibrium (CCE) (Hannan, 1957; Moulin and Vial, 1978)

a distribution $\mu \in \Delta(A)$ is a CCE if for every player i and fixed deviation $a'_i \in A_i$

$$\underbrace{\mathbb{E}_{\mu}[u_i(a)]}_{\text{utility of delegation}} \geq \underbrace{\mathbb{E}_{\mu}[u_i(a'_i, a_{-i})]}_{\text{utility of a fixed action } a'_i}$$

CCE : weak intermediary

Coarse Correlated Equilibrium (CCE) ([Hannan, 1957](#); [Moulin and Vial, 1978](#))

a distribution $\mu \in \Delta(A)$ is a CCE if for every player i and fixed deviation $a'_i \in A_i$

$$\underbrace{\mathbb{E}_{\mu}[u_i(a)]}_{\text{utility of delegation}} \geq \underbrace{\mathbb{E}_{\mu}[u_i(a'_i, a_{-i})]}_{\text{utility of a fixed action } a'_i}$$

contrast with Correlated Equilibrium (CE) ([Aumann, 1987](#)):

- CE captures recommendations rather than delegation
- a player can reconsider a recommendation a_i **after** seeing it
- μ is a CE iff for all i and maps $\sigma_i: A_i \rightarrow A_i$

$$\mathbb{E}_{\mu}[u_i(a)] \geq \mathbb{E}_{\mu}[u_i(\sigma_i(a_i), a_{-i})]$$

IRCP: strong intermediary

Individually Rational Correlated Profile (IRCP) (Myerson, 1991, Chapter 6)

$\mu \in \Delta(A)$ is an IRCP if for every i ,

$$\underbrace{\mathbb{E}_{\mu}[u_i(a)]}_{\text{utility of delegation}} \geq \underbrace{\max_{\nu \in \Delta(A_i)} \min_{a_{-i}} \mathbb{E}_{a_i \sim \nu}[u_i(a_i, a_{-i})]}_{\text{what } i \text{ can guarantee against adversarial play}}$$

IRCP: strong intermediary

Individually Rational Correlated Profile (IRCP) (Myerson, 1991, Chapter 6)

$\mu \in \Delta(A)$ is an IRCP if for every i ,

$$\underbrace{\mathbb{E}_{\mu}[u_i(a)]}_{\text{utility of delegation}} \geq \underbrace{\max_{\nu \in \Delta(A_i)} \min_{a_{-i}} \mathbb{E}_{a_i \sim \nu}[u_i(a_i, a_{-i})]}_{\text{what } i \text{ can guarantee against adversarial play}}$$

- RHS is i 's **maximin guarantee** against adversarial (possibly correlated) play of others

IRCP: strong intermediary

Individually Rational Correlated Profile (IRCP) (Myerson, 1991, Chapter 6)

$\mu \in \Delta(A)$ is an IRCP if for every i ,

$$\underbrace{\mathbb{E}_{\mu}[u_i(a)]}_{\text{utility of delegation}} \geq \underbrace{\max_{\nu \in \Delta(A_i)} \min_{a_{-i}} \mathbb{E}_{a_i \sim \nu}[u_i(a_i, a_{-i})]}_{\text{what } i \text{ can guarantee against adversarial play}}$$

- RHS is i 's **maximin guarantee** against adversarial (possibly correlated) play of others
- chain of inclusions:

$$\text{NE} \subseteq \text{CE} \subseteq \text{CCE} \subseteq \text{IRCP}$$

Unique IRCP

a unique IRCP cannot be mixed

Proposition

If $\text{IRCP}(G)$ is a singleton, then it is a pure strict NE

a unique IRCP cannot be mixed

Proposition

If $\text{IRCP}(G)$ is a singleton, then it is a pure strict NE

Proof:

- IRCP always contains the product of players' maximin strategies $\mu = \mu_1 \times \dots \times \mu_n$

a unique IRCP cannot be mixed

Proposition

If $\text{IRCP}(G)$ is a singleton, then it is a pure strict NE

Proof:

- IRCP always contains the product of players' maximin strategies $\mu = \mu_1 \times \dots \times \mu_n$
- if i switches to her best response b_i to μ_{-i} , then $\delta_{b_i} \times \mu_{-i}$ is again an IRCP

a unique IRCP cannot be mixed

Proposition

If $\text{IRCP}(G)$ is a singleton, then it is a pure strict NE

Proof:

- IRCP always contains the product of players' maximin strategies $\mu = \mu_1 \times \dots \times \mu_n$
- if i switches to her best response b_i to μ_{-i} , then $\delta_{b_i} \times \mu_{-i}$ is again an IRCP
- $\text{IRCP} = \{\mu\} \Rightarrow \mu_i = \delta_{b_i}$ and best response is unique

a unique IRCP cannot be mixed

Proposition

If $\text{IRCP}(G)$ is a singleton, then it is a **pure strict** NE

Proof:

- IRCP always contains the product of players' **maximin strategies** $\mu = \mu_1 \times \dots \times \mu_n$
- if i switches to her best response b_i to μ_{-i} , then $\delta_{b_i} \times \mu_{-i}$ is again an IRCP
- $\text{IRCP} = \{\mu\} \Rightarrow \mu_i = \delta_{b_i}$ and best response is unique
- $\Rightarrow \mu$ is a pure strict Nash equilibrium



a unique IRCP cannot be mixed

Proposition

If $\text{IRCP}(G)$ is a singleton, then it is a pure strict NE

Proof:

- IRCP always contains the product of players' maximin strategies $\mu = \mu_1 \times \dots \times \mu_n$
- if i switches to her best response b_i to μ_{-i} , then $\delta_{b_i} \times \mu_{-i}$ is again an IRCP
- $\text{IRCP} = \{\mu\} \Rightarrow \mu_i = \delta_{b_i}$ and best response is unique
- $\Rightarrow \mu$ is a pure strict Nash equilibrium □

Conclusion: in our search for games with a unique IRCP, we can focus on those with pure strict NE

Dominance is not enough

- dominant profile $\Rightarrow$ unique NE, CE, CCE

Dominance is not enough

- dominant profile $\Rightarrow$ unique NE, CE, CCE
- but IRCP is closed under Pareto improvements

Dominance is not enough

- dominant profile $\Rightarrow$ unique NE, CE, CCE
- but IRCP is closed under Pareto improvements
- $\Rightarrow$ can remain large even with dominance

prisoner's dilemma

	C	D
C	(3, 3)	(0, 4)
D	(4, 0)	(1, 1)

Dominance is not enough

- dominant profile $\Rightarrow$ unique NE, CE, CCE
- but IRCP is closed under Pareto improvements
- $\Rightarrow$ can remain large even with dominance

prisoner's dilemma

	C	D
C	(3, 3)	(0, 4)
D	(4, 0)	(1, 1)

- (D, D) is dominant
- yet (C, C) (and mixtures) are in IRCP

Dominance is not enough

- dominant profile $\Rightarrow$ unique NE, CE, CCE
- but IRCP is closed under Pareto improvements
- $\Rightarrow$ can remain large even with dominance

prisoner's dilemma

	C	D
C	(3, 3)	(0, 4)
D	(4, 0)	(1, 1)

- (D, D) is dominant
- yet (C, C) (and mixtures) are in IRCP

In fact, dominance is orthogonal to IRCP uniqueness

Example: parking enforcement

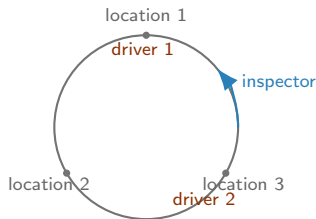
- two drivers choose **park illegally** at one of m locations on a loop or **park legally elsewhere** (cost c)

Example: parking enforcement

- two drivers choose **park illegally** at one of m locations on a loop or **park legally elsewhere** (cost c)
- if exactly one parks illegally $\Rightarrow$ caught for sure (ticket $t > c$)

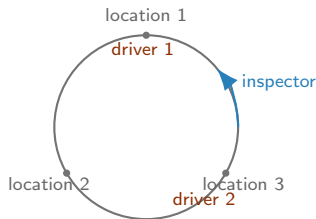
Example: parking enforcement

- two drivers choose **park illegally** at one of m locations on a loop or **park legally elsewhere** (cost c)
- if exactly one parks illegally $\Rightarrow$ caught for sure (ticket $t > c$)
- if both violate, inspection order creates a **strategic race**: only the first found is ticketed
 - ▶ the inspector starts from a uniformly random point and goes counterclockwise



Example: parking enforcement

- two drivers choose **park illegally** at one of m locations on a loop or **park legally elsewhere** (cost c)
- if exactly one parks illegally $\Rightarrow$ caught for sure (ticket $t > c$)
- if both violate, inspection order creates a **strategic race**: only the first found is ticketed
 - ▶ the inspector starts from a uniformly random point and goes counterclockwise

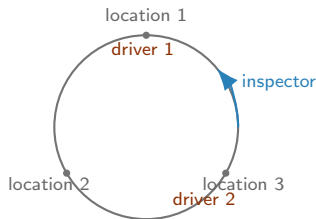


Analysis:

- a violator wants to park right after another violator $\Rightarrow$ caught with probability $1/m$

Example: parking enforcement

- two drivers choose **park illegally** at one of m locations on a loop or **park legally elsewhere** (cost c)
- if exactly one parks illegally $\Rightarrow$ caught for sure (ticket $t > c$)
- if both violate, inspection order creates a **strategic race**: only the first found is ticketed
 - ▶ the inspector starts from a uniformly random point and goes counterclockwise

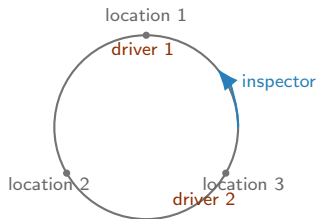


Analysis:

- a violator wants to park right after another violator $\Rightarrow$ caught with probability $1/m$
- $\Rightarrow$ **parking legally** is dominant for $\frac{t}{m} \geq c$

Example: parking enforcement

- two drivers choose **park illegally** at one of m locations on a loop or **park legally elsewhere** (cost c)
- if exactly one parks illegally $\Rightarrow$ caught for sure (ticket $t > c$)
- if both violate, inspection order creates a **strategic race**: only the first found is ticketed
 - ▶ the inspector starts from a uniformly random point and goes counterclockwise

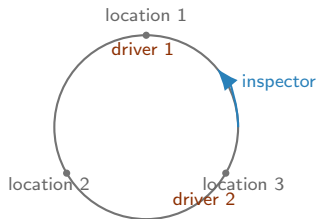


Analysis:

- a violator wants to park right after another violator $\Rightarrow$ caught with probability $1/m$
- $\Rightarrow$ **parking legally** is dominant for $\frac{t}{m} \geq c$
- However, **parking legally** is a unique IRCP for $\frac{t}{2} > c$

Example: parking enforcement

- two drivers choose **park illegally** at one of m locations on a loop or **park legally elsewhere** (cost c)
- if exactly one parks illegally $\Rightarrow$ caught for sure (ticket $t > c$)
- if both violate, inspection order creates a **strategic race**: only the first found is ticketed
 - ▶ the inspector starts from a uniformly random point and goes counterclockwise



Analysis:

- a violator wants to park right after another violator $\Rightarrow$ caught with probability $1/m$
- $\Rightarrow$ **parking legally** is dominant for $\frac{t}{m} \geq c$
- However, **parking legally** is a unique IRCP for $\frac{t}{2} > c$
- **Intuition:** at least one violator will be caught with probability $\geq \frac{1}{2} \Rightarrow$ prefers legal parking

Enforcement games

Enforcement game

A game $H = (A, v)$ is an **enforcement game** if there exists $a^* \in A$ such that

$$v_i(a^*) = 0, \quad v_i(a_i^*, a_{-i}) \geq 0 \quad \forall i, a_{-i}, \quad \sum_i v_i(a) < 0 \quad \forall a \neq a^*.$$

Enforcement games

Enforcement game

A game $H = (A, v)$ is an **enforcement game** if there exists $a^* \in A$ such that

$$v_i(a^*) = 0, \quad v_i(a_i^*, a_{-i}) \geq 0 \quad \forall i, a_{-i}, \quad \sum_i v_i(a) < 0 \quad \forall a \neq a^*.$$

- a^* is **self-enforcing**: each i can guarantee 0 by playing a_i^*

Enforcement games

Enforcement game

A game $H = (A, v)$ is an **enforcement game** if there exists $a^* \in A$ such that

$$v_i(a^*) = 0, \quad v_i(a_i^*, a_{-i}) \geq 0 \quad \forall i, a_{-i}, \quad \sum_i v_i(a) < 0 \quad \forall a \neq a^*.$$

- a^* is **self-enforcing**: each i can guarantee 0 by playing a_i^*
- any other profile is **negative-sum**: someone must drop below their guarantee

Enforcement games

Enforcement game

A game $H = (A, v)$ is an **enforcement game** if there exists $a^* \in A$ such that

$$v_i(a^*) = 0, \quad v_i(a_i^*, a_{-i}) \geq 0 \quad \forall i, a_{-i}, \quad \sum_i v_i(a) < 0 \quad \forall a \neq a^*.$$

- a^* is **self-enforcing**: each i can guarantee 0 by playing a_i^*
- any other profile is **negative-sum**: someone must drop below their guarantee

Lemma

a^* is a unique IRCP

Enforcement games

Enforcement game

A game $H = (A, v)$ is an **enforcement game** if there exists $a^* \in A$ such that

$$v_i(a^*) = 0, \quad v_i(a_i^*, a_{-i}) \geq 0 \quad \forall i, a_{-i}, \quad \sum_i v_i(a) < 0 \quad \forall a \neq a^*.$$

- a^* is **self-enforcing**: each i can guarantee 0 by playing a_i^*
- any other profile is **negative-sum**: someone must drop below their guarantee

Lemma

a^* is a unique IRCP

Proof:

- if $\mu(A \setminus \{a^*\}) > 0$, then expected welfare < 0
- $\Rightarrow$ some player violates IR constraint



Characterization of games with unique IRCP

Theorem 1

A game G has a unique IRCP $\iff G$ is **affinely equivalent** to an enforcement game

- $G = (A, u)$ and $H = (A, v)$ are affine equivalent if $v_i = \gamma_i(u_i + \beta_i)$ with $\gamma_i > 0$ and $\beta_i \in \mathbb{R}$

Characterization of games with unique IRCP

Theorem 1

A game G has a unique IRCP $\iff G$ is **affinely equivalent** to an enforcement game

- $G = (A, u)$ and $H = (A, v)$ are affine equivalent if $v_i = \gamma_i(u_i + \beta_i)$ with $\gamma_i > 0$ and $\beta_i \in \mathbb{R}$

Proof sketch:

- **“if”**: Lemma & IRCP respecting affine equivalence

Characterization of games with unique IRCP

Theorem 1

A game G has a unique IRCP $\iff G$ is **affinely equivalent** to an enforcement game

- $G = (A, u)$ and $H = (A, v)$ are affine equivalent if $v_i = \gamma_i(u_i + \beta_i)$ with $\gamma_i > 0$ and $\beta_i \in \mathbb{R}$

Proof sketch:

- “**if**”: Lemma & IRCP respecting affine equivalence
- “**only if**”: unique IRCP = $\{a^*\} \Rightarrow$

$$\max_{\mu \in \text{IRCP}} \mu(A \setminus \{a^*\}) = 0$$

Characterization of games with unique IRCP

Theorem 1

A game G has a unique IRCP $\iff G$ is **affinely equivalent** to an enforcement game

- $G = (A, u)$ and $H = (A, v)$ are affine equivalent if $v_i = \gamma_i(u_i + \beta_i)$ with $\gamma_i > 0$ and $\beta_i \in \mathbb{R}$

Proof sketch:

- “**if**”: Lemma & IRCP respecting affine equivalence
- “**only if**”: unique IRCP = $\{a^*\} \Rightarrow$

$$\max_{\mu \in \text{IRCP}} \mu(A \setminus \{a^*\}) = 0$$

the dual to this LP gives γ_i and β_i

□

Unique CCE

Weak delegation: invariances matter

- CCE is invariant to payoff shifts that depend only on **others'** actions
 - ▶ $\Leftarrow$ CCE compares utility differences across own actions

Weak delegation: invariances matter

- CCE is invariant to payoff shifts that depend only on **others'** actions
 - ▶ $\Leftarrow$ CCE compares utility differences across own actions

Strategic equivalence

$G = (A, u)$ and $H = (A, v)$ are strategically equivalent if

$$v_i(a_i, a_{-i}) = \gamma_i(u_i(a_i, a_{-i}) + \beta_i(a_{-i})), \quad \gamma_i > 0$$

Weak delegation: invariances matter

- CCE is invariant to payoff shifts that depend only on **others'** actions
 - ▶ $\Leftrightarrow$ CCE compares utility differences across own actions

Strategic equivalence

$G = (A, u)$ and $H = (A, v)$ are strategically equivalent if

$$v_i(a_i, a_{-i}) = \gamma_i(u_i(a_i, a_{-i}) + \beta_i(a_{-i})), \quad \gamma_i > 0$$

- In general, $\text{IRCP}(G) \neq \text{IRCP}(H)$ since $\beta_i(a_{-i})$ can affect the ability of others to punish i

Weak delegation: invariances matter

- CCE is invariant to payoff shifts that depend only on **others'** actions
 - ▶ $\Leftarrow$ CCE compares utility differences across own actions

Strategic equivalence

$G = (A, u)$ and $H = (A, v)$ are strategically equivalent if

$$v_i(a_i, a_{-i}) = \gamma_i(u_i(a_i, a_{-i}) + \beta_i(a_{-i})), \quad \gamma_i > 0$$

- In general, $\text{IRCP}(G) \neq \text{IRCP}(H)$ since $\beta_i(a_{-i})$ can affect the ability of others to punish i
- However, $\text{CCE}(G) = \text{CCE}(H)$

Weak delegation: invariances matter

- CCE is invariant to payoff shifts that depend only on **others'** actions
 - ▶ $\Leftarrow$ CCE compares utility differences across own actions

Strategic equivalence

$G = (A, u)$ and $H = (A, v)$ are strategically equivalent if

$$v_i(a_i, a_{-i}) = \gamma_i(u_i(a_i, a_{-i}) + \beta_i(a_{-i})), \quad \gamma_i > 0$$

- In general, $\text{IRCP}(G) \neq \text{IRCP}(H)$ since $\beta_i(a_{-i})$ can affect the ability of others to punish i
- However, $\text{CCE}(G) = \text{CCE}(H)$
- $\Rightarrow$ CCE is **unique and pure** for any G strategically equivalent to an enforcement game

Weak delegation: invariances matter

- CCE is invariant to payoff shifts that depend only on **others'** actions
 - ▶ $\Leftarrow$ CCE compares utility differences across own actions

Strategic equivalence

$G = (A, u)$ and $H = (A, v)$ are strategically equivalent if

$$v_i(a_i, a_{-i}) = \gamma_i(u_i(a_i, a_{-i}) + \beta_i(a_{-i})), \quad \gamma_i > 0$$

- In general, $\text{IRCP}(G) \neq \text{IRCP}(H)$ since $\beta_i(a_{-i})$ can affect the ability of others to punish i
- However, $\text{CCE}(G) = \text{CCE}(H)$
- $\Rightarrow$ CCE is **unique and pure** for any G strategically equivalent to an enforcement game

We first focus on pure CCE and discuss mixed CCE later

Characterization of unique pure CCE

Theorem 2

A game G has a **unique and pure** CCE $\iff G$ is strategically equivalent to an enforcement game

Characterization of unique pure CCE

Theorem 2

A game G has a **unique and pure** CCE $\iff G$ is strategically equivalent to an enforcement game

- a connection between games with unique CCE and unique IRCP

Characterization of unique pure CCE

Theorem 2

A game G has a **unique and pure** CCE $\iff G$ is strategically equivalent to an enforcement game

- a connection between games with unique CCE and unique IRCP
- furthermore, a^* is the unique CCE of G iff a^* is the unique IRCP of the transformed game

$$v_i(a) = u_i(a) - u_i(a_i^*, a_{-i}),$$

that is, one can always take $\beta_i(a_{-i}) = -u_i(a_i^*, a_{-i})$ in strategic equivalence

Characterization of unique pure CCE

Theorem 2

A game G has a **unique and pure** CCE $\iff G$ is strategically equivalent to an enforcement game

- a connection between games with unique CCE and unique IRCP
- furthermore, a^* is the unique CCE of G iff a^* is the unique IRCP of the transformed game

$$v_i(a) = u_i(a) - u_i(a_i^*, a_{-i}),$$

that is, one can always take $\beta_i(a_{-i}) = -u_i(a_i^*, a_{-i})$ in strategic equivalence

- $\Rightarrow$ an easy-to check criterion for unique CCE

Testing for unique pure CCE

Criterion for uniqueness

a^* is a unique CCE iff there exist weights $\gamma_i > 0$ such that for all $a \neq a^*$,

$$\Phi_{a^*}(a) = \sum_i \gamma_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right) < 0$$

For symmetric games, can take $\gamma_i \equiv 1$

Testing for unique pure CCE

Criterion for uniqueness

a^* is a unique CCE iff there exist weights $\gamma_i > 0$ such that for all $a \neq a^*$,

$$\Phi_{a^*}(a) = \sum_i \gamma_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right) < 0$$

For symmetric games, can take $\gamma_i \equiv 1$

Interpretation:

- Φ is a **local potential** for deviations from a^*

Testing for unique pure CCE

Criterion for uniqueness

a^* is a unique CCE iff there exist weights $\gamma_i > 0$ such that for all $a \neq a^*$,

$$\Phi_{a^*}(a) = \sum_i \gamma_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right) < 0$$

For symmetric games, can take $\gamma_i \equiv 1$

Interpretation:

- Φ is a **local potential** for deviations from a^*
- indeed, for unilateral deviations $a = (a_i, a_{-i}^*)$,

$$\Phi_{a^*}(a_i, a_{-i}^*) = \gamma_i \left(u_i(a_i, a_{-i}^*) - u_i(a^*) \right)$$

Testing for unique pure CCE

Criterion for uniqueness

a^* is a unique CCE iff there exist weights $\gamma_i > 0$ such that for all $a \neq a^*$,

$$\Phi_{a^*}(a) = \sum_i \gamma_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right) < 0$$

For symmetric games, can take $\gamma_i \equiv 1$

Interpretation:

- Φ is a **local potential** for deviations from a^*
- indeed, for unilateral deviations $a = (a_i, a_{-i}^*)$,

$$\Phi_{a^*}(a_i, a_{-i}^*) = \gamma_i \left(u_i(a_i, a_{-i}^*) - u_i(a^*) \right)$$

- thus $\Phi_{a^*}(a_i, a_{-i}^*) < 0 \Leftrightarrow a^*$ is a strict Nash equilibrium

Testing for unique pure CCE

Criterion for uniqueness

a^* is a unique CCE iff there exist weights $\gamma_i > 0$ such that for all $a \neq a^*$,

$$\Phi_{a^*}(a) = \sum_i \gamma_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right) < 0$$

For symmetric games, can take $\gamma_i \equiv 1$

Interpretation:

- Φ is a **local potential** for deviations from a^*
- indeed, for unilateral deviations $a = (a_i, a_{-i}^*)$,

$$\Phi_{a^*}(a_i, a_{-i}^*) = \gamma_i (u_i(a_i, a_{-i}^*) - u_i(a^*))$$

- thus $\Phi_{a^*}(a_i, a_{-i}^*) < 0 \Leftrightarrow a^*$ is a strict Nash equilibrium
- uniqueness of CCE additionally requires **multilateral deviations** to be negative-sum in Φ

A simple sufficient condition for unique CCE

$$\Phi_{a^*}(a) = \sum_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right)$$

Idea: for strictly concave functions, coordinate-wise maximum $\Rightarrow$ global maximum

A simple sufficient condition for unique CCE

$$\Phi_{a^*}(a) = \sum_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right)$$

Idea: for strictly concave functions, coordinate-wise maximum $\Rightarrow$ global maximum

Sufficient condition for uniqueness

Suppose $A_i \subset \mathbb{R}^d$ are convex. Then a NE a^* is a unique CCE if Φ_{a^*} is strictly concave

A simple sufficient condition for unique CCE

$$\Phi_{a^*}(a) = \sum_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right)$$

Idea: for strictly concave functions, coordinate-wise maximum $\Rightarrow$ global maximum

Sufficient condition for uniqueness

Suppose $A_i \subset \mathbb{R}^d$ are convex. Then a NE a^* is a unique CCE if Φ_{a^*} is strictly concave

Example: strictly socially concave games ([Even-Dar, Mansour, and Nadav, 2009](#)):

- A_i is compact and convex
- each $u_i(a_i, a_{-i})$ is concave in a_i and convex in a_{-i}
- utilitarian welfare $\sum_i u_i(a)$ is **strictly concave** in a

A simple sufficient condition for unique CCE

$$\Phi_{a^*}(a) = \sum_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right)$$

Idea: for strictly concave functions, coordinate-wise maximum $\Rightarrow$ global maximum

Sufficient condition for uniqueness

Suppose $A_i \subset \mathbb{R}^d$ are convex. Then a NE a^* is a unique CCE if Φ_{a^*} is strictly concave

Example: strictly socially concave games ([Even-Dar, Mansour, and Nadav, 2009](#)):

- A_i is compact and convex
- each $u_i(a_i, a_{-i})$ is concave in a_i and convex in a_{-i}
- utilitarian welfare $\sum_i u_i(a)$ is **strictly concave** in a

Corollary: CCE is unique in strictly socially concave games ([Hart and Mas-Colell, 2015](#))

A simple sufficient condition for unique CCE

$$\Phi_{a^*}(a) = \sum_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right)$$

Idea: for strictly concave functions, coordinate-wise maximum $\Rightarrow$ global maximum

Sufficient condition for uniqueness

Suppose $A_i \subset \mathbb{R}^d$ are convex. Then a NE a^* is a unique CCE if Φ_{a^*} is strictly concave

Example: strictly socially concave games ([Even-Dar, Mansour, and Nadav, 2009](#)):

- A_i is compact and convex
- each $u_i(a_i, a_{-i})$ is concave in a_i and convex in a_{-i}
- utilitarian welfare $\sum_i u_i(a)$ is **strictly concave** in a

Corollary: CCE is unique in strictly socially concave games ([Hart and Mas-Colell, 2015](#))

- this class includes linear Cournot and Tullock contests with strictly convex costs
- but there are economically relevant examples beyond

Application: contests & potential games

Contests: why uniqueness matters

- standard model: players exert costly effort a_i to win a prize

Contests: why uniqueness matters

- standard model: players exert costly effort a_i to win a prize
- utilities (two-player contest):

$$u_i(a) = v_i p_i(a) - c_i(a_i), \quad p_1(a) + p_2(a) = 1$$

Contests: why uniqueness matters

- standard model: players exert costly effort a_i to win a prize
- utilities (two-player contest):

$$u_i(a) = v_i p_i(a) - c_i(a_i), \quad p_1(a) + p_2(a) = 1$$

- collusion concern: players share the prize, save effort

Contests: why uniqueness matters

- standard model: players exert costly effort a_i to win a prize
- utilities (two-player contest):

$$u_i(a) = v_i p_i(a) - c_i(a_i), \quad p_1(a) + p_2(a) = 1$$

- collusion concern: players share the prize, save effort
- uniqueness of CCE $\Rightarrow$ no scope for collusion

Unique CCE in two-player contests

Proposition

If a contest admits a **strict pure** NE a^* , then a^* is the **unique** CCE

Unique CCE in two-player contests

Proposition

If a contest admits a **strict pure** NE a^* , then a^* is the **unique** CCE

Proof sketch:

- choose weights $\gamma_i = 1/v_i$ in the local potential and use $p_1 + p_2 \equiv 1$ and get

$$\Phi_{a^*}(a) = \frac{1}{v_1} \left(u_1(a_1, a_2^*) - u_1(a^*) \right) + \frac{1}{v_2} \left(u_2(a_1^*, a_2) - u_2(a^*) \right)$$

Unique CCE in two-player contests

Proposition

If a contest admits a **strict pure** NE a^* , then a^* is the **unique** CCE

Proof sketch:

- choose weights $\gamma_i = 1/v_i$ in the local potential and use $p_1 + p_2 \equiv 1$ and get

$$\Phi_{a^*}(a) = \frac{1}{v_1} \left(u_1(a_1, a_2^*) - u_1(a^*) \right) + \frac{1}{v_2} \left(u_2(a_1^*, a_2) - u_2(a^*) \right)$$

- $\Phi < 0$ whenever i deviates from a_i^*



Unique CCE in two-player contests

Proposition

If a contest admits a **strict pure** NE a^* , then a^* is the **unique** CCE

Proof sketch:

- choose weights $\gamma_i = 1/v_i$ in the local potential and use $p_1 + p_2 \equiv 1$ and get

$$\Phi_{a^*}(a) = \frac{1}{v_1} \left(u_1(a_1, a_2^*) - u_1(a^*) \right) + \frac{1}{v_2} \left(u_2(a_1^*, a_2) - u_2(a^*) \right)$$

- $\Phi < 0$ whenever i deviates from a_i^*

□

Remark: separable Φ is another class where coordinate-wise maximum $\Rightarrow$ global maximum

Ratio-based contests

- symmetric ratio-based contests: $p_1(a) = f\left(\frac{a_1}{a_2}\right)$ and $f(t) + f(1/t) = 1$
 - ▶ **Example:** $f(x) = \frac{x}{1+x}$ (Tullock), $f(x) = \mathbb{1}_{\{x>1\}}$ (1st price auction)

Ratio-based contests

- symmetric ratio-based contests: $p_1(a) = f\left(\frac{a_1}{a_2}\right)$ and $f(t) + f(1/t) = 1$
 - ▶ **Example:** $f(x) = \frac{x}{1+x}$ (Tullock), $f(x) = \mathbb{1}_{\{x>1\}}$ (1st price auction)
- assume linear cost $c_i(a_i) = a_i$ and $v_1 = v_2 = 1$

Ratio-based contests

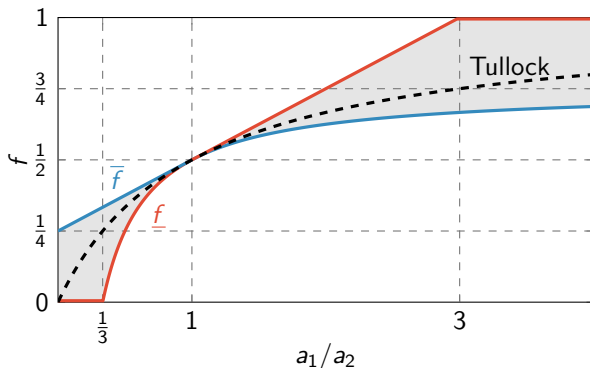
- symmetric ratio-based contests: $p_1(a) = f\left(\frac{a_1}{a_2}\right)$ and $f(t) + f(1/t) = 1$
 - ▶ **Example:** $f(x) = \frac{x}{1+x}$ (Tullock), $f(x) = \mathbb{1}_{\{x>1\}}$ (1st price auction)
- assume linear cost $c_i(a_i) = a_i$ and $v_1 = v_2 = 1$
- for Tullock, effort $1/4$ is a strict NE ($\Rightarrow$ unique CCE)

Ratio-based contests

- symmetric ratio-based contests: $p_1(a) = f\left(\frac{a_1}{a_2}\right)$ and $f(t) + f(1/t) = 1$
 - ▶ **Example:** $f(x) = \frac{x}{1+x}$ (Tullock), $f(x) = \mathbb{1}_{\{x>1\}}$ (1st price auction)
- assume linear cost $c_i(a_i) = a_i$ and $v_1 = v_2 = 1$
- for Tullock, effort $1/4$ is a strict NE ($\Rightarrow$ unique CCE)
- consider all symmetric ratio-based contests with linear costs where effort $1/4$ is a unique CCE

Ratio-based contests

- symmetric ratio-based contests: $p_1(a) = f\left(\frac{a_1}{a_2}\right)$ and $f(t) + f(1/t) = 1$
 - ▶ **Example:** $f(x) = \frac{x}{1+x}$ (Tullock), $f(x) = \mathbb{1}_{\{x>1\}}$ (1st price auction)
- assume linear cost $c_i(a_i) = a_i$ and $v_1 = v_2 = 1$
- for Tullock, effort $1/4$ is a strict NE ($\Rightarrow$ unique CCE)
- consider all symmetric ratio-based contests with linear costs where effort $1/4$ is a unique CCE
- a contest is in this class $\Leftrightarrow f$ lies between $\bar{f}$ and $\underline{f}$



Potential games

- a **potential game** is one strategically equivalent to $u_i \equiv P$ (Monderer and Shapley, 1996)
- Neyman (1997): convex $A_i \subseteq \mathbb{R}^d$, P is strictly concave and smooth $\Rightarrow$ unique CE $a^* = \arg \max P$

Potential games

- a **potential game** is one strategically equivalent to $u_i \equiv P$ (Monderer and Shapley, 1996)
- Neyman (1997): convex $A_i \subseteq \mathbb{R}^d$, P is strictly concave and smooth $\Rightarrow$ unique CE $a^* = \arg \max P$

Corollary (necessary condition for CCE uniqueness)

under Neyman (1997)'s assumptions, if CCE is unique, then the Hessian H of P at a^* satisfies

$$H - \frac{1}{2}\text{diag}(H) \preceq 0$$

Potential games

- a **potential game** is one strategically equivalent to $u_i \equiv P$ (Monderer and Shapley, 1996)
- Neyman (1997): convex $A_i \subseteq \mathbb{R}^d$, P is strictly concave and smooth $\Rightarrow$ unique CE $a^* = \arg \max P$

Corollary (necessary condition for CCE uniqueness)

under Neyman (1997)'s assumptions, if CCE is unique, then the Hessian H of P at a^* satisfies

$$H - \frac{1}{2}\text{diag}(H) \preceq 0$$

- a diagonal dominance condition limiting cross-effects

Potential games

- a **potential game** is one strategically equivalent to $u_i \equiv P$ (Monderer and Shapley, 1996)
- Neyman (1997): convex $A_i \subseteq \mathbb{R}^d$, P is strictly concave and smooth $\Rightarrow$ unique CE $a^* = \arg \max P$

Corollary (necessary condition for CCE uniqueness)

under Neyman (1997)'s assumptions, if CCE is unique, then the Hessian H of P at a^* satisfies

$$H - \frac{1}{2} \text{diag}(H) \preceq 0$$

- a diagonal dominance condition limiting cross-effects
- for $P(a) = \langle a, Q a \rangle - \sum_i c_i(a_i)$ with convex c_i , the condition with $\prec$ is also sufficient

Potential games

- a **potential game** is one strategically equivalent to $u_i \equiv P$ (Monderer and Shapley, 1996)
- Neyman (1997): convex $A_i \subseteq \mathbb{R}^d$, P is strictly concave and smooth $\Rightarrow$ unique CE $a^* = \arg \max P$

Corollary (necessary condition for CCE uniqueness)

under Neyman (1997)'s assumptions, if CCE is unique, then the Hessian H of P at a^* satisfies

$$H - \frac{1}{2} \text{diag}(H) \preceq 0$$

- a diagonal dominance condition limiting cross-effects
- for $P(a) = \langle a, Q a \rangle - \sum_i c_i(a_i)$ with convex c_i , the condition with $\prec$ is also sufficient
- separates environments with unique CE from those with unique CCE

Unique mixed CCE

Example: 2×2 games

matching pennies		
	H	T
H	$(1, -1)$	$(-1, 1)$
T	$(-1, 1)$	$(1, -1)$

Example: 2×2 games

matching pennies		
	H	T
H	$(1, -1)$	$(-1, 1)$
T	$(-1, 1)$	$(1, -1)$

- the unique NE is also unique CCE

Example: 2×2 games

matching pennies		
	H	T
H	$(1, -1)$	$(-1, 1)$
T	$(-1, 1)$	$(1, -1)$

- the unique NE is also unique CCE
- true for all **games of matching pennies type**: 2×2 games with a unique NE which is mixed

Example: 2×2 games

matching pennies		
	H	T
H	$(1, -1)$	$(-1, 1)$
T	$(-1, 1)$	$(1, -1)$

- the unique NE is also unique CCE
- true for all **games of matching pennies type**: 2×2 games with a unique NE which is mixed

Conclusion: in contrast to IRCP, unique CCE can be mixed

Unique CCE is “almost pure”

Proposition

If $\text{CCE}(G)$ is unique, then either

- it is a **pure** NE, or
- exactly two players mix over two actions each, and the induced 2×2 game is of **matching-pennies type**.

Unique CCE is “almost pure”

Proposition

If $\text{CCE}(G)$ is unique, then either

- it is a **pure** NE, or
 - exactly two players mix over two actions each, and the induced 2×2 game is of **matching-pennies type**.
-
- in **symmetric** games, uniqueness $\Rightarrow$ purity

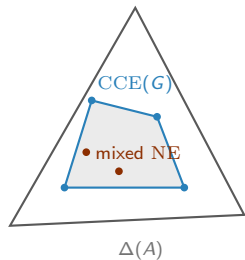
Unique CCE is “almost pure”

Proposition

If $\text{CCE}(G)$ is unique, then either

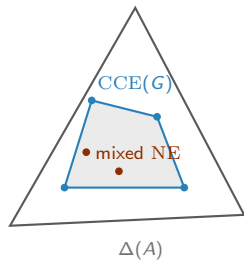
- it is a **pure** NE, or
 - exactly two players mix over two actions each, and the induced 2×2 game is of **matching-pennies type**.
-
- in **symmetric** games, uniqueness $\Rightarrow$ purity
 - mixed uniqueness is essentially a two-player artifact

Why richer mixing is ruled out?



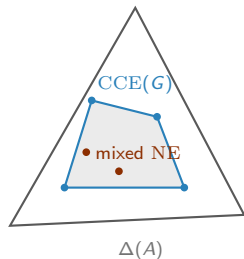
- $CCE(G)$ is a polytope, a convex hull of its **extreme points**

Why richer mixing is ruled out?



- $CCE(G)$ is a polytope, a convex hull of its **extreme points**
- if $CCE(G) = \{\nu\}$, then ν is a quasi-strict NE and (trivially) an extreme point

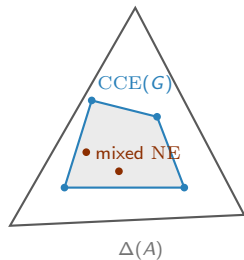
Why richer mixing is ruled out?



- $\text{CCE}(G)$ is a polytope, a convex hull of its **extreme points**
- if $\text{CCE}(G) = \{\nu\}$, then ν is a quasi-strict NE and (trivially) an extreme point
- quasi-strict mixed NE with support sizes $k_1, \dots, k_m$ satisfies

$$\# \text{support} = \prod_{i=1}^m k_i, \quad \# \text{active constraints} \leq 1 + \sum_{i=1}^m k_i$$

Why richer mixing is ruled out?



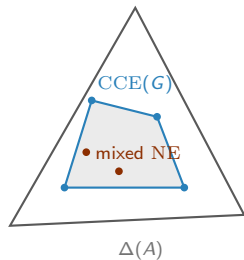
- $CCE(G)$ is a polytope, a convex hull of its **extreme points**
- if $CCE(G) = \{\nu\}$, then ν is a quasi-strict NE and (trivially) an extreme point
- quasi-strict mixed NE with support sizes $k_1, \dots, k_m$ satisfies

$$\# \text{support} = \prod_{i=1}^m k_i, \quad \# \text{active constraints} \leq 1 + \sum_{i=1}^m k_i$$

- extreme point $\Rightarrow$

$$\prod_{i=1}^m k_i \leq 1 + \sum_{i=1}^m k_i$$

Why richer mixing is ruled out?



- $\text{CCE}(G)$ is a polytope, a convex hull of its **extreme points**
- if $\text{CCE}(G) = \{\nu\}$, then ν is a quasi-strict NE and (trivially) an extreme point
- quasi-strict mixed NE with support sizes $k_1, \dots, k_m$ satisfies

$$\# \text{support} = \prod_{i=1}^m k_i, \quad \# \text{active constraints} \leq 1 + \sum_{i=1}^m k_i$$

- extreme point $\Rightarrow$

$$\prod_{i=1}^m k_i \leq 1 + \sum_{i=1}^m k_i$$

Proposition

A quasi-strict NE is extreme in CCE $\iff$ it is pure or 2×2 mixing

Extensions

When does delegation just select among Nash?

- a more permissive benchmark for when delegation doesn't expand outcomes:

$$\text{delegation outcomes} = \text{conv}(\text{NE}(G))$$

- the intermediary only selects which NE to play

When does delegation just select among Nash?

- a more permissive benchmark for when delegation doesn't expand outcomes:

$$\text{delegation outcomes} = \text{conv}(\text{NE}(G))$$

- the intermediary only selects which NE to play

Proposition

For a generic game G ,

either $\text{IRCP}(G)$ is a **singleton** OR $\text{conv}(\text{NE}(G))$ is a **proper subset** of $\text{IRCP}(G)$

When does delegation just select among Nash?

- a more permissive benchmark for when delegation doesn't expand outcomes:

$$\text{delegation outcomes} = \text{conv}(\text{NE}(G))$$

- the intermediary only selects which NE to play

Proposition

For a generic game G ,

either $\text{IRCP}(G)$ is a **singleton** OR $\text{conv}(\text{NE}(G))$ is a **proper subset** of $\text{IRCP}(G)$

- the more permissive benchmark doesn't bring anything new

Why $\text{conv}(\text{NE}) \neq \text{IRCP}$ unless a singleton?

- genericity $\Rightarrow$ a finite set of quasi-strict NE ([Harsanyi, 1973](#))

Why $\text{conv}(\text{NE}) \neq \text{IRCP}$ unless a singleton?

- genericity $\Rightarrow$ a finite set of quasi-strict NE ([Harsanyi, 1973](#))
- assume for contradiction $\text{conv}(\text{NE}(G)) = \text{IRCP}(G)$ and $\text{IRCP}(G)$ has ≥ 2 points

Why $\text{conv}(\text{NE}) \neq \text{IRCP}$ unless a singleton?

- genericity $\Rightarrow$ a finite set of quasi-strict NE ([Harsanyi, 1973](#))
- assume for contradiction $\text{conv}(\text{NE}(G)) = \text{IRCP}(G)$ and $\text{IRCP}(G)$ has ≥ 2 points
- extreme points of $\text{IRCP}(G)$ are NE

Why $\text{conv}(\text{NE}) \neq \text{IRCP}$ unless a singleton?

- genericity $\Rightarrow$ a finite set of quasi-strict NE ([Harsanyi, 1973](#))
- assume for contradiction $\text{conv}(\text{NE}(G)) = \text{IRCP}(G)$ and $\text{IRCP}(G)$ has ≥ 2 points
- extreme points of $\text{IRCP}(G)$ are NE
- quasi-strictness + extremality $\Rightarrow$ [pure](#) (2×2 mixing is ruled out for IRCP)

Why $\text{conv}(\text{NE}) \neq \text{IRCP}$ unless a singleton?

- genericity $\Rightarrow$ a finite set of quasi-strict NE (Harsanyi, 1973)
- assume for contradiction $\text{conv}(\text{NE}(G)) = \text{IRCP}(G)$ and $\text{IRCP}(G)$ has ≥ 2 points
- extreme points of $\text{IRCP}(G)$ are NE
- quasi-strictness + extremality $\Rightarrow$ pure (2×2 mixing is ruled out for IRCP)
- For generic G , k strict pure NE $\Rightarrow \exists k - 1$ mixed NE (Gül, Pearce, and Stacchetti, 1993)

Why $\text{conv}(\text{NE}) \neq \text{IRCP}$ unless a singleton?

- genericity $\Rightarrow$ a finite set of quasi-strict NE (Harsanyi, 1973)
- assume for contradiction $\text{conv}(\text{NE}(G)) = \text{IRCP}(G)$ and $\text{IRCP}(G)$ has ≥ 2 points
- extreme points of $\text{IRCP}(G)$ are NE
- quasi-strictness + extremality $\Rightarrow$ pure (2×2 mixing is ruled out for IRCP)
- For generic G , k strict pure NE $\Rightarrow \exists k - 1$ mixed NE (Gül, Pearce, and Stacchetti, 1993)
- A mixed NE requires *indifferences* on its support, which cannot be generated as a convex combination of strict pure NE

When is $\text{conv}(\text{NE}) = \text{CE}$ or CCE?

Open question

When is $\text{conv}(\text{NE}) = \text{CE}$ or CCE?

there are environments with multiple NE where correlation does not expand outcomes

- potential games: $\text{conv}(\text{NE}) = \text{CE}$ ([Neyman, 1997](#); [Ui, 2008](#); [Cao, Tan, and Zhou, 2025](#))
- zero-sum games: $\text{conv}(\text{NE}) = \text{CE}$ ([Forges, 1990](#); [Viossat, 2006](#))
- auctions: $\text{conv}(\text{NE}) = \text{CE}$ ([Feldman, Lucier, and Nisan, 2016](#))
- non-atomic convex potential games: $\text{conv}(\text{NE}) = \text{CCE}$ ([Koessler, Scarsini, and Tomala, 2025](#))

can this happen generically?

Open question

When is $\text{conv}(\text{NE}) = \text{CE}$ or CCE?

there are environments with multiple NE where correlation does not expand outcomes

- potential games: $\text{conv}(\text{NE}) = \text{CE}$ (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025)
- zero-sum games: $\text{conv}(\text{NE}) = \text{CE}$ (Forges, 1990; Viossat, 2006)
- auctions: $\text{conv}(\text{NE}) = \text{CE}$ (Feldman, Lucier, and Nisan, 2016)
- non-atomic convex potential games: $\text{conv}(\text{NE}) = \text{CCE}$ (Koessler, Scarsini, and Tomala, 2025)

can this happen generically?

“No” in 2×2 games (Calvó-Armengol, 2006) and, more generally, two-person games (Cripps, 1995)

Take-away

- unique delegation outcomes in broad classes of games
- **strong intermediary (IRCP):** uniqueness $\Leftrightarrow$ enforcement up to affine equivalence
- **weak intermediary (CCE):** uniqueness $\Leftrightarrow$ enforcement up to strategic equivalence
- **design takeaway:** collusion-proofness is a tractable and permissive constraint
it is enough that multilateral deviations create **negative-sum externalities**

- unique delegation outcomes in broad classes of games
- **strong intermediary (IRCP)**: uniqueness $\Leftrightarrow$ enforcement up to affine equivalence
- **weak intermediary (CCE)**: uniqueness $\Leftrightarrow$ enforcement up to strategic equivalence
- **design takeaway**: collusion-proofness is a tractable and permissive constraint
it is enough that multilateral deviations create **negative-sum externalities**

Thank you!

References I

- Mete Şeref Ahunbay and Martin Bichler. On the uniqueness of bayesian coarse correlated equilibria in standard first-price and all-pay auctions. In *Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 2491–2537. SIAM, 2025.
- Stephanie Assad, Robert Clark, Daniel Ershov, and Lei Xu. Algorithmic pricing and competition: Empirical evidence from the german retail gasoline market. *Journal of Political Economy*, 132(3): 723–771, 2024.
- Robert J Aumann. Subjectivity and correlation in randomized strategies. *Journal of mathematical Economics*, 1(1):67–96, 1974.
- Robert J Aumann. Correlated equilibrium as an expression of bayesian rationality. *Econometrica: Journal of the Econometric Society*, pages 1–18, 1987.
- Robert J Aumann and Lloyd S Shapley. Long-term competition—a game-theoretic analysis. In *reprinted in Essays in Game Theory: In Honor of Michael Maschler (ed. N. Megiddo)*, Springer, 1994, pp. 1–15. 1976.
- Imre Bárány. Fair distribution protocols or how the players replace fortune. *Mathematics of Operations Research*, 17(2):327–340, 1992.

References II

- Lucas Baudin, Marco Scarsini, and Xavier Venel. Strategic behavior and no-regret learning in queueing systems. *arXiv preprint arXiv:2302.03614*, 2023.
- Elchanan Ben-Porath. Correlation without mediation: Expanding the set of equilibrium outcomes by “cheap” pre-play procedures. *Journal of Economic Theory*, 80(1):108–122, 1998.
- Jean-Pierre Benoit, Vijay Krishna, et al. Finitely repeated games. 1984.
- Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolo, and Sergio Pastorello. Artificial intelligence, algorithmic pricing, and collusion. *American Economic Review*, 110(10):3267–3297, 2020.
- Antoni Calvó-Armengol. The set of correlated equilibria of 2x2 games. *mimeo*, 2006.
- Zhigang Cao, Zhibin Tan, and Jinchuan Zhou. Correlated equilibrium of games with concave potential functions. *Operations Research Letters*, 60:107241, 2025.
- M Cripps. Extreme correlated and Nash equilibria in two-person games, 1995.
- Eyal Even-Dar, Yishay Mansour, and Uri Nadav. On the convergence of regret minimization dynamics in concave games. In *Proceedings of the forty-first annual ACM symposium on Theory of computing*, pages 523–532, 2009.

References III

- Michal Feldman, Brendan Lucier, and Noam Nisan. Correlated and coarse equilibria of single-item auctions. In *Web and Internet Economics: 12th International Conference, WINE 2016, Montreal, Canada, December 11-14, 2016, Proceedings 12*, pages 131–144. Springer, 2016.
- Sara Fish, Yannai A Gonczarowski, and Ran I Shorrer. Algorithmic collusion by large language models. *arXiv preprint arXiv:2404.00806*, 7, 2024.
- Françoise Forges. Correlated equilibrium in two-person zero-sum games. *Econometrica (1986-1998)*, 58(2):515, 1990.
- Dean P Foster and Rakesh V Vohra. Calibrated learning and correlated equilibrium. *Games and Economic Behavior*, 21(1-2):40–55, 1997.
- Drew Fudenberg and David K Levine. Conditional universal consistency. *Games and Economic Behavior*, 29(1-2):104–130, 1999.
- Drew Fudenberg and Eric Maskin. The folk theorem in repeated games with discounting or with incomplete information. *Econometrica*, 54(3):533–554, 1986.
- Dino Gerardi. Unmediated communication in games with complete and incomplete information. *Journal of Economic Theory*, 114(1):104–131, 2004.

References IV

- Faruk Gül, David Pearce, and Ennio Stacchetti. A bound on the proportion of pure strategy equilibria in generic games. *Mathematics of Operations Research*, 18(3):548–552, 1993.
- James Hannan. Approximation to bayes risk in repeated play. *Contributions to the Theory of Games*, 3(2):97–139, 1957.
- John C Harsanyi. Oddness of the number of equilibrium points: a new proof. *International Journal of Game Theory*, 2:235–250, 1973.
- Sergiu Hart and Andreu Mas-Colell. A simple adaptive procedure leading to correlated equilibrium. *Econometrica*, 68(5):1127–1150, 2000.
- Sergiu Hart and Andreu Mas-Colell. Markets, correlation, and regret-matching. *Games and Economic Behavior*, 93:42–58, 2015.
- Navin Kartik, Francesco Squintani, and Katrin Tinn. Information revelation in constant-sum games: Elections and beyond. *arXiv preprint arXiv:2406.17084*, 2024.
- Deniz Kattwinkel, Axel Niemeyer, Justus Preusser, and Alexander Winter. Mechanisms without transfers for fully biased agents. In *Proceedings of the 23rd ACM Conference on Economics and Computation*, pages 332–333, 2022.

References V

- Timo Klein. Autonomous algorithmic collusion: Q-learning under sequential pricing. *The RAND Journal of Economics*, 52(3):538–558, 2021.
- Frédéric Koessler, Marco Scarsini, and Tristan Tomala. Correlated equilibria in large anonymous bayesian games. *Mathematics of Operations Research*, 50(3):2157–2174, 2025.
- Ehud Lehrer and Sylvain Sorin. One-shot public mediated talk. *Games and Economic Behavior*, 20(2): 131–148, 1997.
- Dov Monderer and Lloyd S Shapley. Potential games. *Games and economic behavior*, 14(1):124–143, 1996.
- Hervé Moulin and J P Vial. Strategically zero-sum games: the class of games whose completely mixed equilibria cannot be improved upon. *International Journal of Game Theory*, 7:201–221, 1978.
- Roger B Myerson. *Game Theory: Analysis of Conflict*. Harvard University Press, 1991.
- Uri Nadav and Georgios Piliouras. No regret learning in oligopolies: Cournot vs. Bertrand. In *International Symposium on Algorithmic Game Theory*, pages 300–311. Springer, 2010.
- Abraham Neyman. Correlated equilibrium and potential games. *International Journal of Game Theory*, 26:223–227, 1997.

References VI

- Takashi Ui. Correlated equilibrium and concave games. *International Journal of Game Theory*, 37:1–13, 2008.
- Yannick Viossat. The geometry of Nash equilibria and correlated equilibria and a generalization of zero-sum games. Technical report, SSE/EFI Working Paper Series in Economics and Finance, 2006.